

IoT Water Quality Monitoring System

Syed Affan Ali • Seven parameters. One connected view.

01 / PROBLEM STATEMENT

Water quality needs a broader view

Manual, isolated checks make it difficult to follow changes over time. Multiple water properties need to be collected together and viewed remotely.

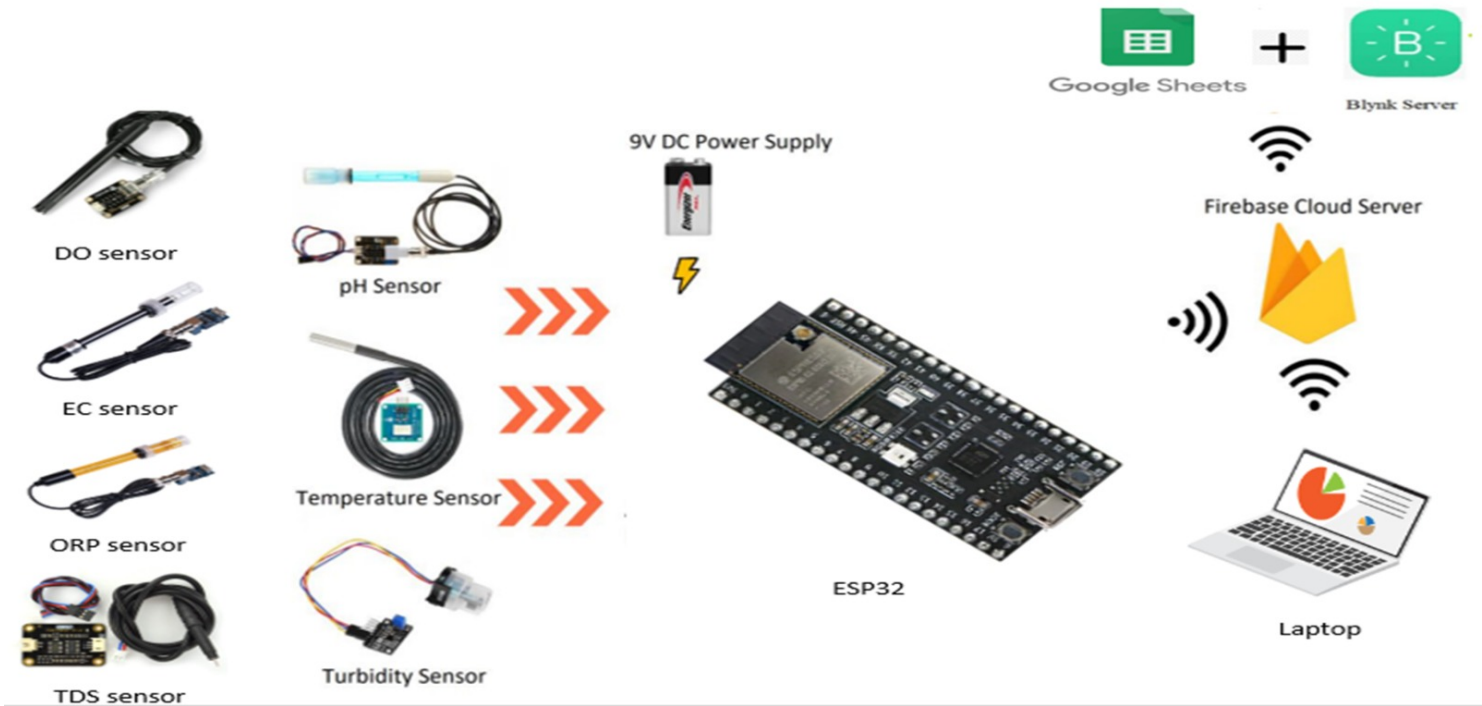
02 / PROJECT OBJECTIVE

Connect measurements to insight

Build an ESP32 prototype that acquires seven sensor readings, publishes them to Blynk, and combines them into a custom Water Quality Index.

03 / SYSTEM ARCHITECTURE

From sensor probes to a mobile dashboard



7 measured parameters

Temperature • Turbidity • pH • EC • ORP • Dissolved oxygen • TDS

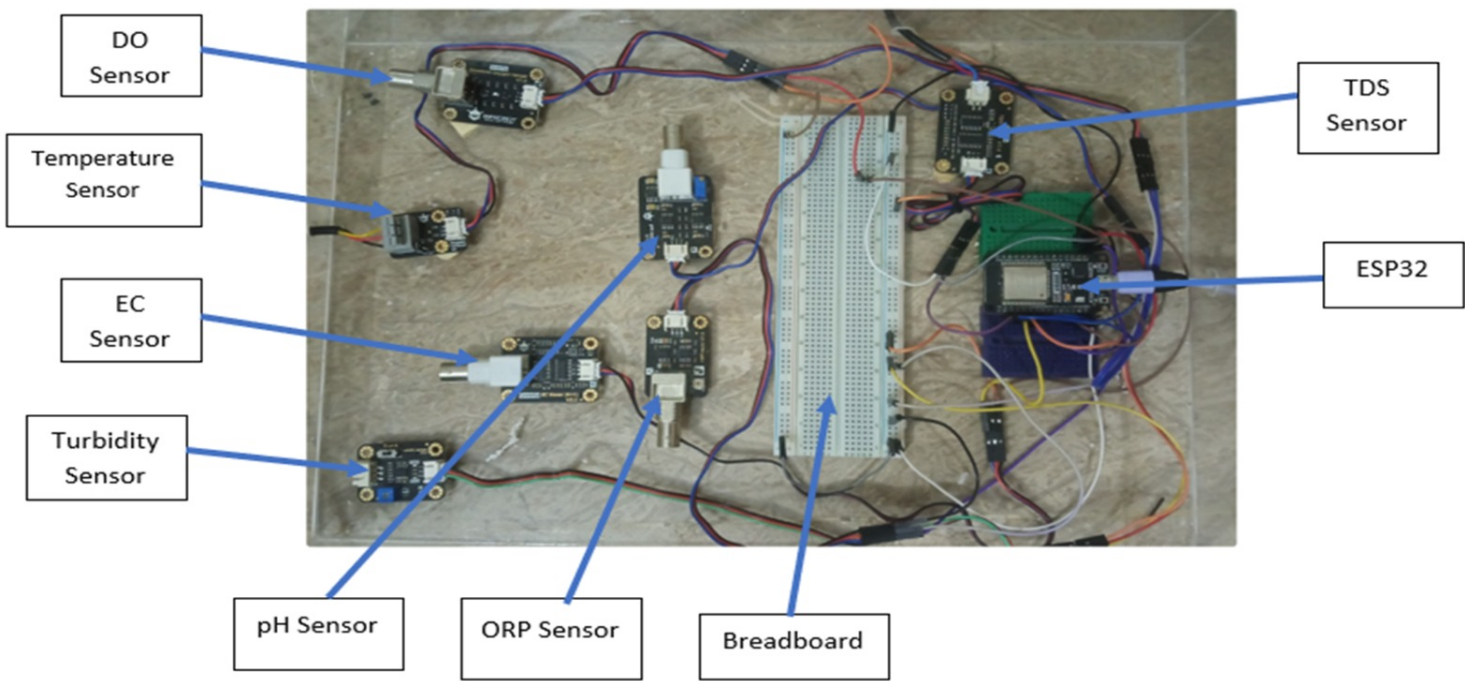
Acquisition → conversion → WQI

Blynk: remote measurement gauges.
Google Sheets: recorded data shown in the project evidence.

Firestore is shown in the architecture; its integration is not verified in the supplied code.

04 / BUILT HARDWARE

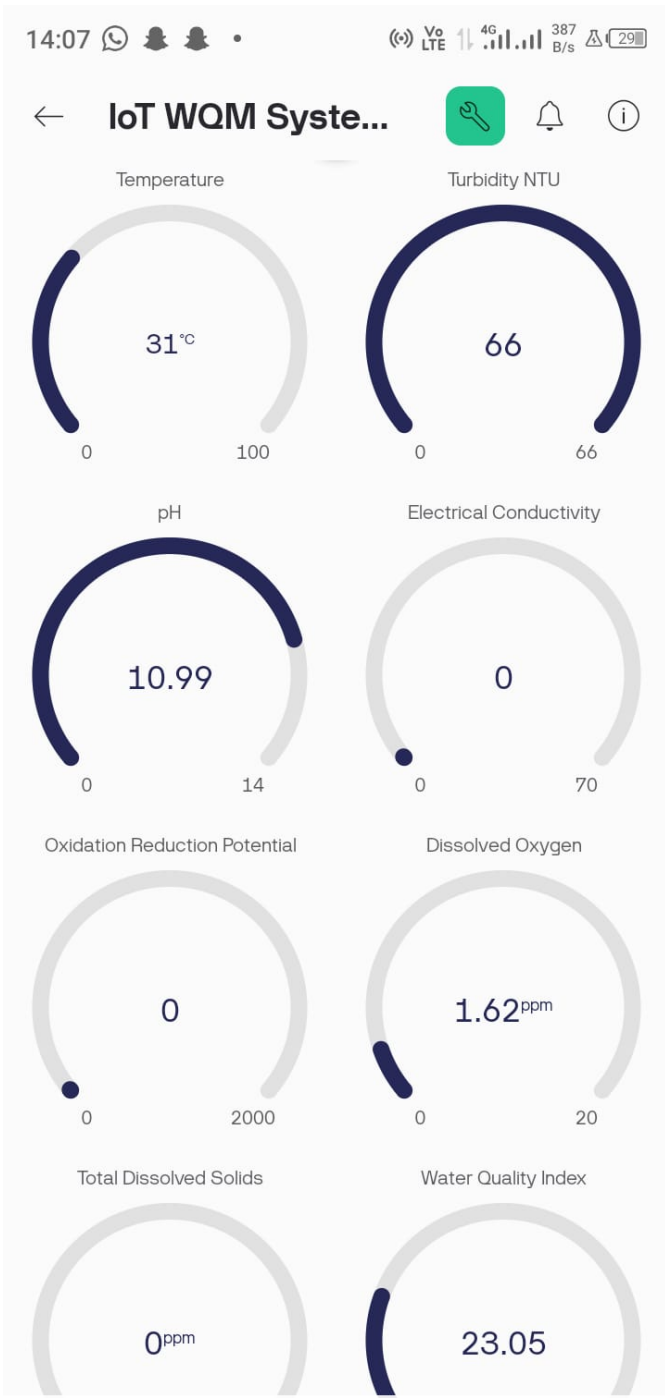
Seven sensor interfaces connected to ESP32



Assembled breadboard prototype with labeled sensor modules.

05 / REMOTE VIEW

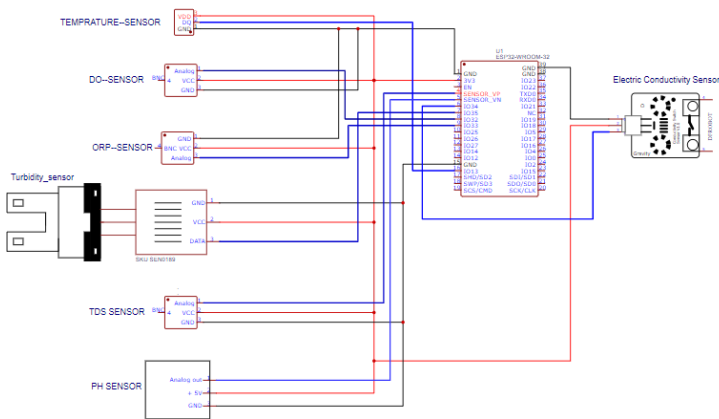
Blynk dashboard



Seven readings + custom WQI

06 / CIRCUIT DESIGN

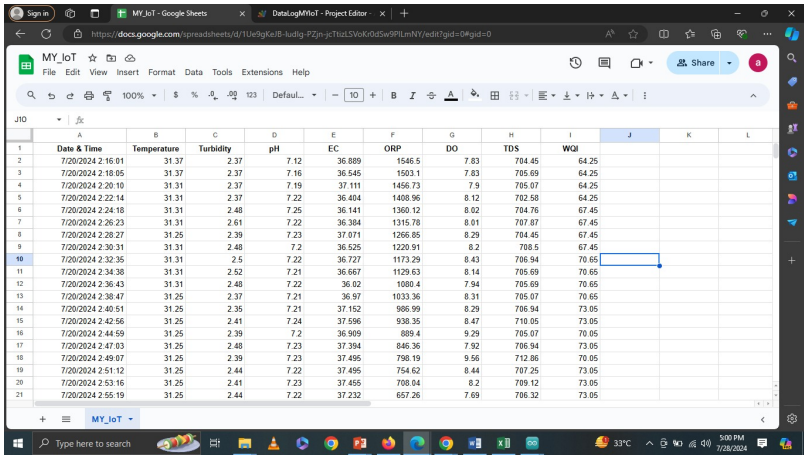
Sensor connections



ESP32 wiring diagram

07 / DATA RECORDS

Google Sheets logging



Timestamped parameters + WQI

WHAT I ACHIEVED

Integrated seven sensors • Implemented conversions and filtering • Published eight Blynk values • Calculated a custom WQI • Demonstrated logged measurements